SECURITY



Caliptra Subsystem 2.2: USB 2.0 and OCP Streaming Boot

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Agenda

Subsystem 2.2 configurations

USB 2.0 streaming boot

Virtual hub: two devices

Runtime protocol examples

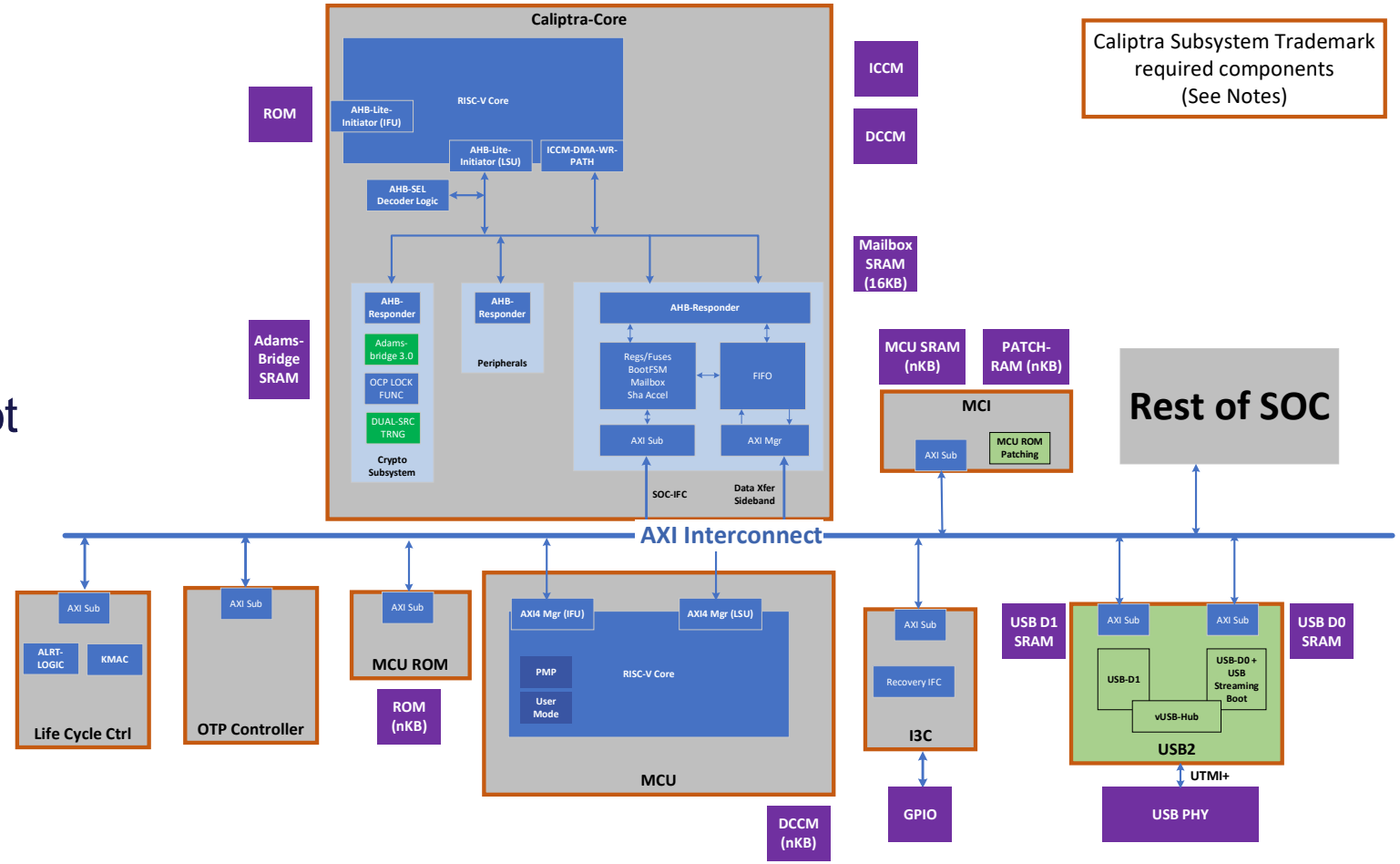
I3C and AXI compatibility

Caliptra: Root of Trust

Measurement and identity

Authentication of every image

Caliptra-owned streaming boot



USB expands streaming boot

I3C Streaming Boot

Existing OCP option

Deployed today

Still supported

AXI Streaming Boot

Integration-specific option

On-die delivery

Still supported

USB 2.0 Streaming Boot

New hardware option

Native OCP streaming boot

Device 0, EP0 provides recovery

USB standardized

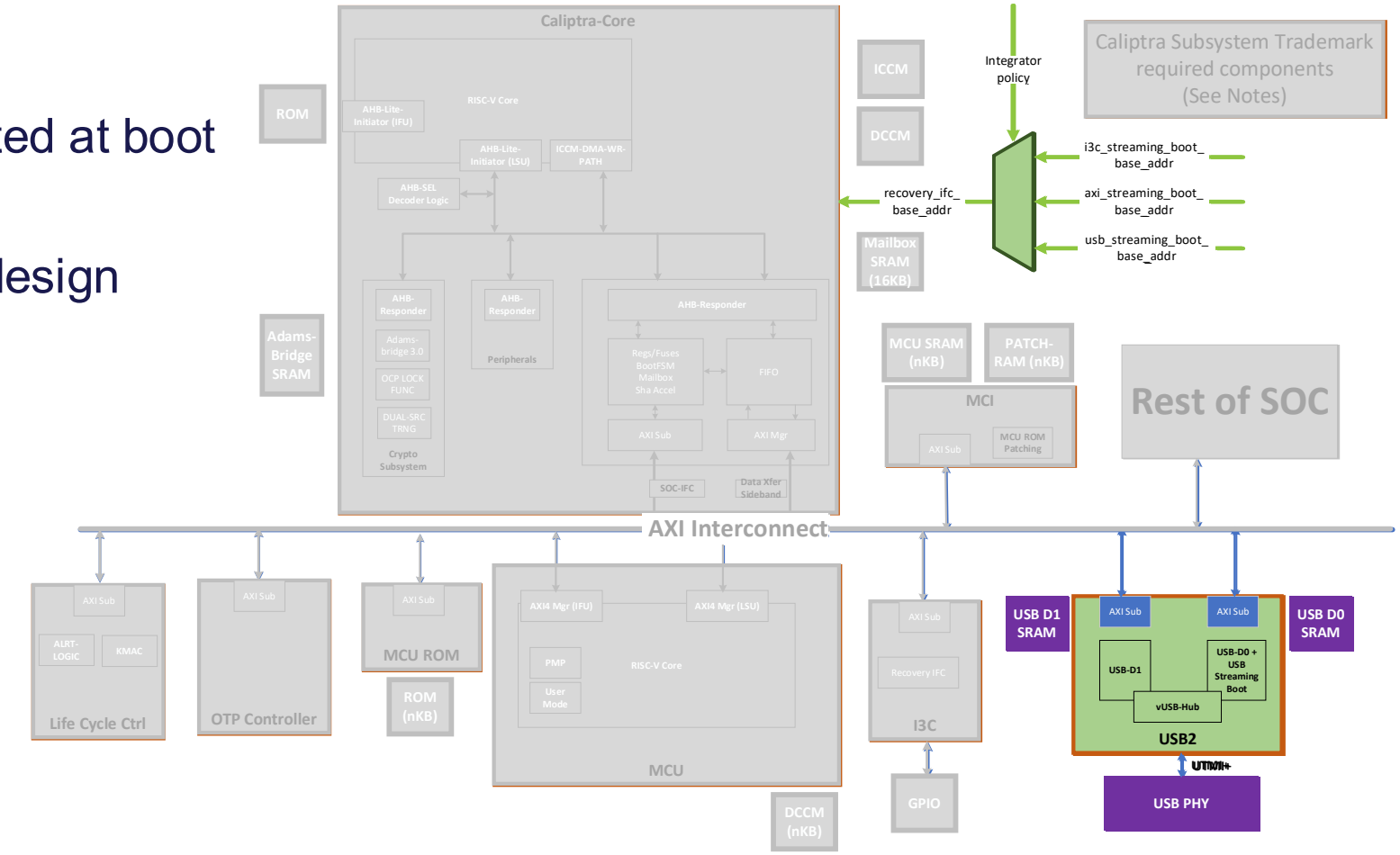
Multiple transport options supported

Streaming Boot Configuration

Streaming Boot Engine selected at boot

I3C, AXI, USB coexist in the design

Flexible runtime flows



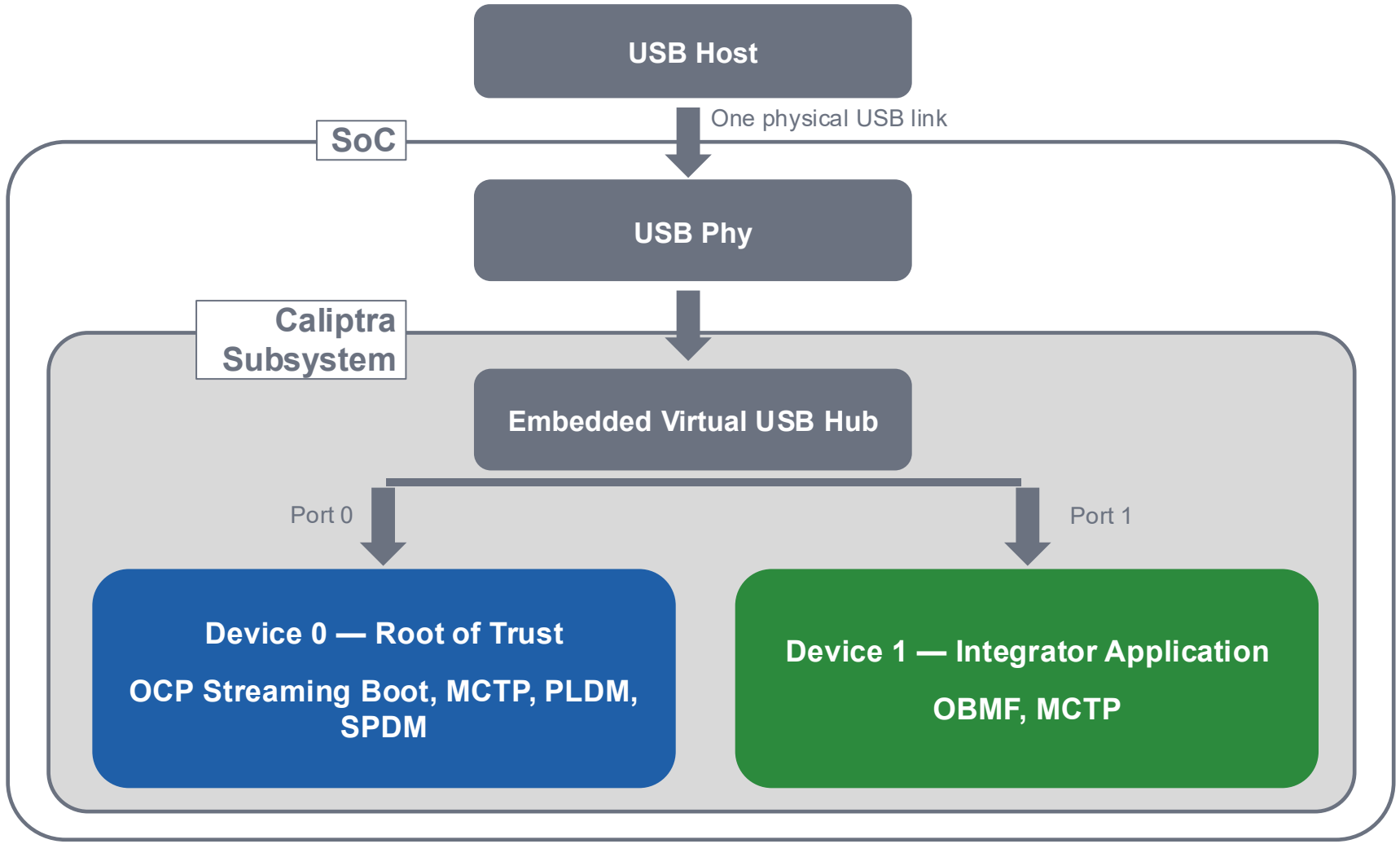
Why USB

1. **One familiar external interface**
   Ubiquitous, hot-pluggable, high-bandwidth
2. **Simpler programming models**
   Standard enumeration and endpoints
3. **Modular ownership**
   Separate owners, separate functions
4. **Fewer pins, less power**
   Less custom interface logic

Companion talk

USB to Rule Them All: One Manageability Interface for AI Platforms

Virtual Hub: One link, two devices



Single USB Phy (outside Caliptra)

Standard hub enumeration

Per-device resources

USB compound device microarchitecture

Single USB PHY interface

- UTMI or ULPI

SYNC & MUX routes packets to handlers

HW FSM handles all USB hub requests

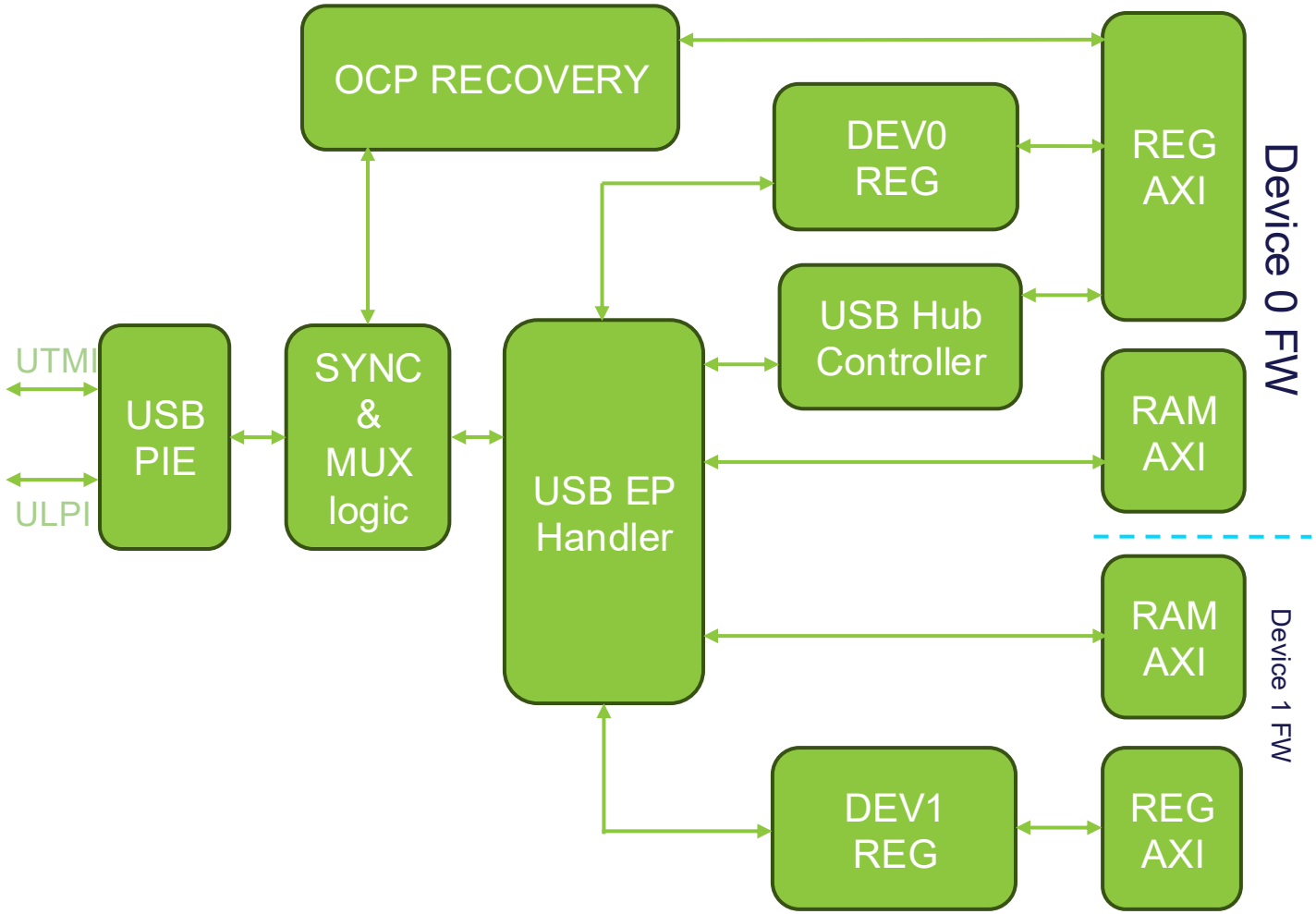
- FW can override descriptor data

SRAM contains endpoint buffers

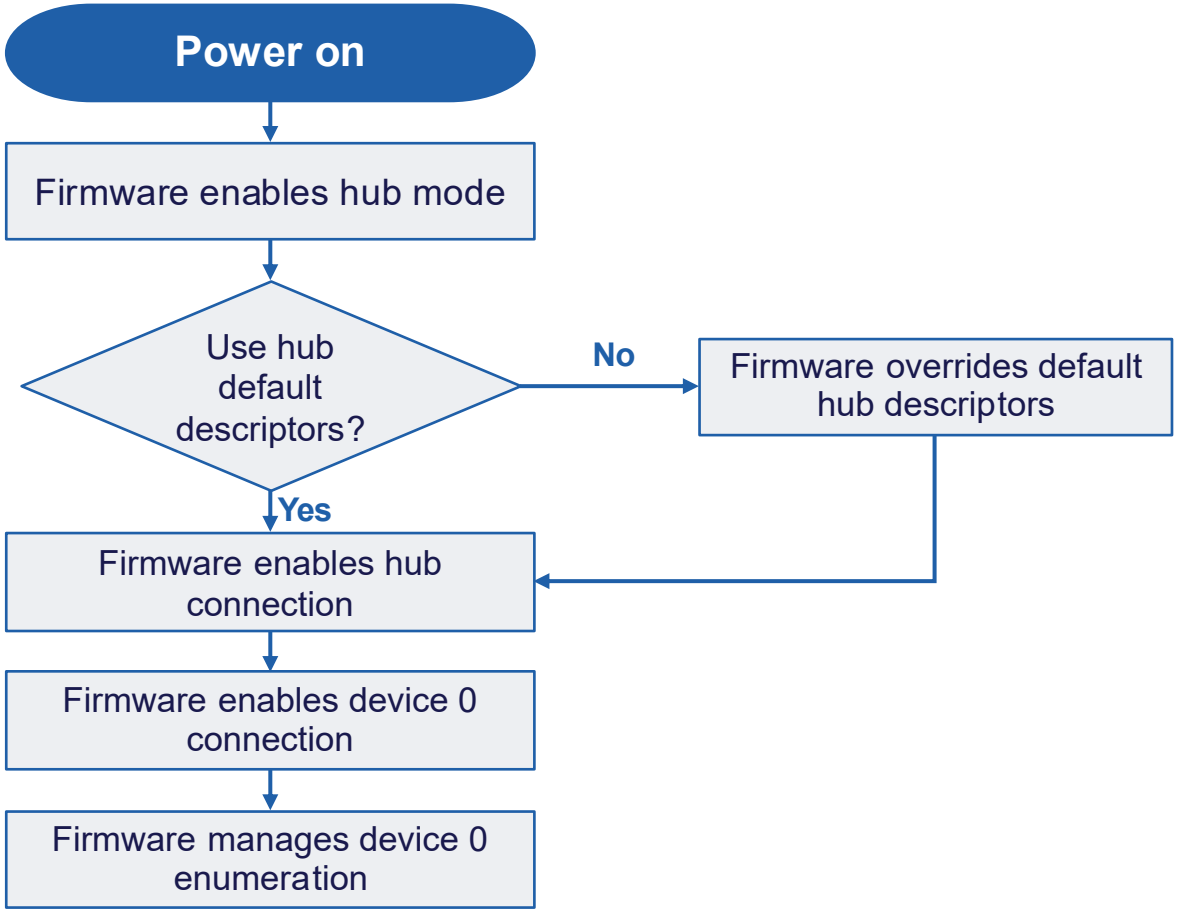
- Dedicated AXI ports

Device 0 and Device 1

- Same programming interface
- Independent operation



USB compound device programming



Independent ownership

Device 0: Root-of-trust

OCP Recovery at boot

Runtime USB after boot

Example: MCTP

Device 1: Integrator application

SoC-defined descriptors

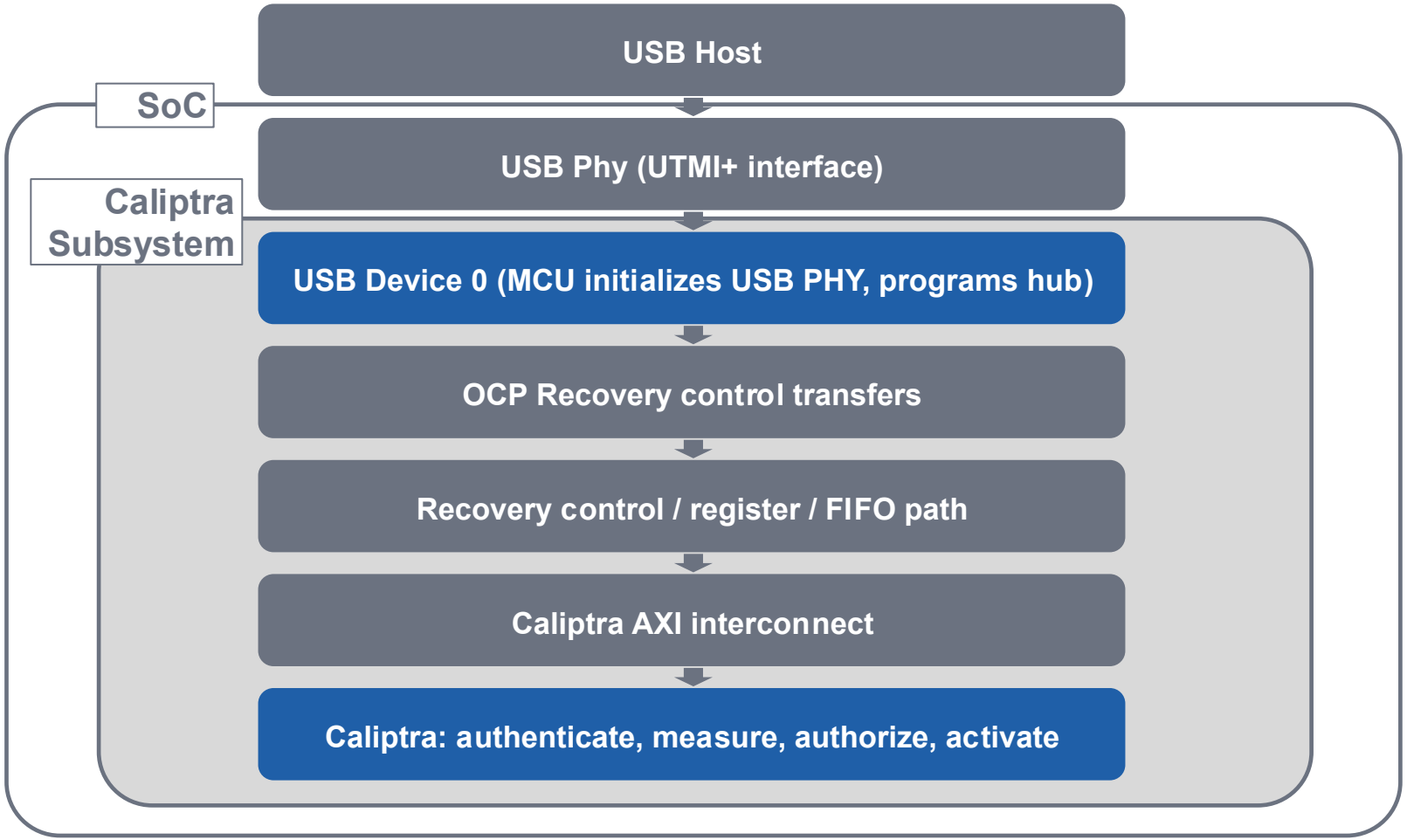
OBMF

MCTP

Vendor-specific channels

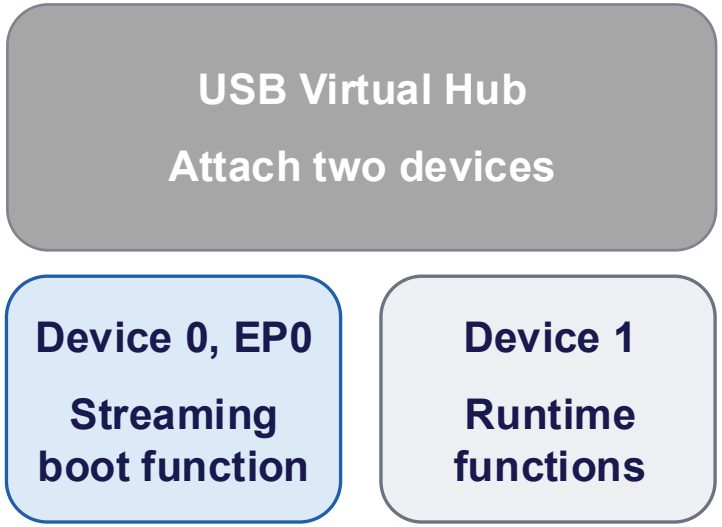
Application protocols are independent of Caliptra recovery

USB recovery path



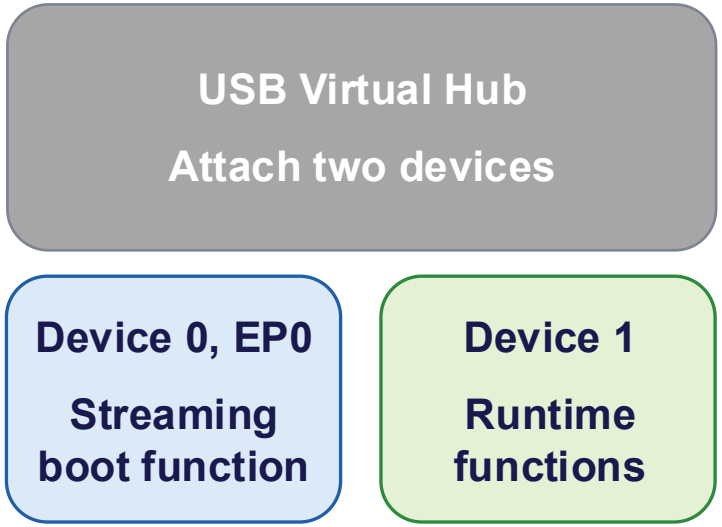
Boot sequence

- Power on, Device 0 mode
- MCU configures, enables embedded hub
- MCU wakes Caliptra
- Caliptra finds OCP Recovery
- Images stream over Device 0
- Caliptra, MCU, SoC runtimes load



Runtime function attachment

- Streaming boot completes
- Caliptra, MCU, SoC running
- MCU signals the SoC
- Device 1 connects
- MCU resets Device 0
- Device 0 re-enumerates, runtime only
- Device 0 exposes MCTP, SPDM, PLDM
- Device 1 exposes MCTP, OBMF, etc.



Integrator model

Programming responsibilities

MCU owns hub and Device 0

SoC owns Device 1

Program resources before connect

Isolate each device (Reset, disconnect, transfer handling)

Compatibility

I3C still supported

AXI still supported

Caliptra 2.2 adds USB option

Secure boot plus runtime services

Summary

Native USB streaming boot

Hub with two owned devices

Recovery at boot, firmware-managed protocols later

Fully independent device ownership

Call to Action

- Join the validation effort!
  - https://github.com/chipsalliance/caliptra-ss/tree/main
- Integrate Caliptra Subsystem 2.2 into your SoC
  - https://github.com/chipsalliance/caliptra-ss/releases (v2.2 coming soon)
- Join the community calls
  - https://github.com/chipsalliance/Caliptra/blob/main/README.md#discussion-channels

Thank You!

